

ASSIGNMENT-11

Exercise 1: Read Count with HISAT2 Alignments

For htseq data:

- **Generate PCA, and PCC plots using vst normalisation.**

To prepare the RNA-seq count data for exploratory analyses, we applied a Variance Stabilizing Transformation (VST) using the script PCA_PCC_using_vst_log_data.R. VST converts raw counts into values where the variance is roughly constant across both low- and high-expression genes. This reduces noise from highly variable low-count genes and produces a matrix that is well-suited for PCA, clustering, and correlation analyses.

Note: For HISAT2 alignment, raw read counts were generated using two different quantification programs. The merged count matrices are:

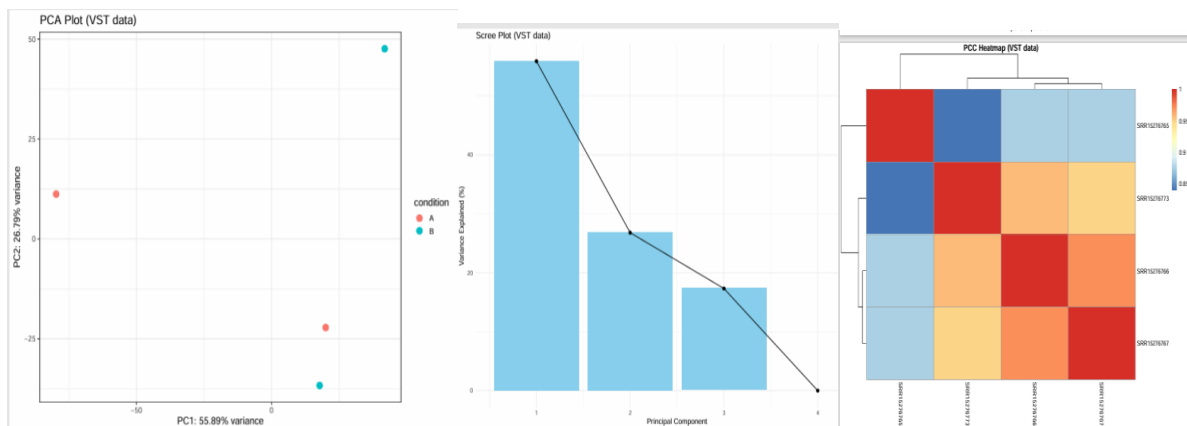
- raw_reads_hisat_feature_count.txt (featureCounts output)
- raw_reads_hisat_htseq.txt (HTSeq output)

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_htseq$ head -5 raw_reads_hisat_htseq.txt
Geneid  SRR15276765  SRR15276766  SRR15276767  SRR15276773
PF3D7_0100100  7      91      106      77
PF3D7_0100200  9      25      10      15
PF3D7_0100300  0      0       4       0
PF3D7_0100400  1      58      95      114
```

```
# -----
# 1. Import read count matrix
# -----
counts <- read.table("/home/ibab/NGS/assignment_11/for_HISAT2/for_htseq/raw_reads_hisat_htseq.txt", header = TRUE, row.names = 1, sep = "\t")

# Sample info
samples <- data.frame(
  row.names = colnames(counts),
  condition = c("A", "A", "B", "B") # adjust groups
)
```

Rscript PCA_PCC_using_vst_log_data.R



- **Generate PCA and PCC plots using TPM normalisation.**

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11$ head -4 Pfalciparum.genome.gtf
M76611 PlasmDB transcript 3 33 . + . transcript_id "rna_malmito_rna_16:rRNA-1"; gene_id "malmito_rna_16:rRNA"
M76611 PlasmDB exon 3 33 . + . transcript_id "rna_malmito_rna_16:rRNA-1"; gene_id "malmito_rna_16:rRNA";
M76611 PlasmDB transcript 34 71 . + . transcript_id "rna_malmito_rna_20:rRNA-1"; gene_id "malmito_rna_20:rRNA"
M76611 PlasmDB exon 34 71 . + . transcript_id "rna_malmito_rna_20:rRNA-1"; gene_id "malmito_rna_20:rRNA";
```

```
awk -F"\t" '$3=="transcript" && $0 ~ /PF3D7/ {
    # Attribute field is $9
    if (match($9,/gene_id "([^"]+)"/,a)) {
        gene=a[1]
        len_kb = ($5 - $4 + 1) / 1000
        print gene "\t" len_kb
    }
}' Pfalciparum.genome.gtf > transcript_length_kb_fixed.txt
```

Parameters	Purpose
awk	Runs the awk program for text processing.
-F"\t"	Sets the field separator to a tab (\t), so \$1...\$9 correctly refer to tab-delimited columns of the GTF.
\$3=="transcript"	Condition: process only lines where the 3rd column (feature column in GTF) equals transcript.
\$0 ~ /PF3D7/	Condition: process only lines where the entire line (\$0) contains the string PF3D7.
if (match(\$9,/gene_id "([^"]+)"/,a))	Searches 9th column (the attributes field) for pattern gene_id "...". - match() finds the pattern ([""]+) captures everything inside the quotes - a[1] stores the captured gene ID.
gene=a[1]	Stores the extracted gene ID into a variable gene.
len_kb = (\$5 - \$4 + 1) / 1000	Calculates transcript length in kilobases: (end position - start position + 1) / 1000.
print gene "\t" len_kb	Prints gene ID and length (kb) separated by a tab.
Pfalciparum.genome.gtf	The input GTF file containing Plasmodium falciparum gene annotations.
> transcript_length_kb_fixed.txt	Redirects output to new file named transcript_length_kb_fixed.txt.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11$ wc -l transcript_length_kb_fixed.txt
5680 transcript_length_kb_fixed.txt
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11$ head -5 transcript_length_kb_fixed.txt
PF3D7_0100100 7.617
PF3D7_0100200 1.226
PF3D7_0100300 4.141
PF3D7_0100400 1.274
PF3D7_0100500 0.112
```

```
sort -k1,1 raw_reads_hisat_htseq.txt > raw_reads.sorted.txt
sort -k1,1 transcript_length_kb_fixed.txt > transcript_length.sorted.txt
join -t $'\t' -1 1 -2 1 raw_reads.sorted.txt transcript_length.sorted.txt \
> raw_reads_with_length.txt
```

Command	Key Option	Purpose
sort -k1,1 raw_reads_hisat_htseq.txt > raw_reads.sorted.txt	-k1,1	Sort raw reads by Gene ID (1st column).
sort -k1,1 transcript_length_kb_fixed.txt > transcript_length.sorted.txt	-k1,1	Sort length file by Gene ID.
join -t \$'\t' -1 1 -2 1 raw_reads.sorted.txt transcript_length.sorted.txt > raw_reads_with_length.txt	-t \$'\t'	Use tab as separator.
	-1 1	Join on 1st column of file1.
	-2 1	Join on 1st column of file2.

Add header and move length column right after GeneID.

```
awk 'BEGIN { OFS="\t"; print
"Gene_ID","Gene_length","SRR15276765","SRR15276766","SRR15276767","SRR1527677
3" } { print $1,$NF,$2,$3,$4,$5 }' raw_reads_with_length.txt >
raw_reads_with_length_reordered.txt
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11$ wc -l raw_reads_with_length_reordered.txt
5681 raw_reads_with_length_reordered.txt
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11$ head -5 raw_reads_with_length_reordered.txt
Gene_ID Gene_length SRR15276765 SRR15276766 SRR15276767 SRR15276773
PF3D7_0100100 7.617 7 91 106 77
PF3D7_0100200 1.226 9 25 10 15
PF3D7_0100300 4.141 0 0 4 0
PF3D7_0100400 1.274 1 58 95 114
```

TPM normalizes for both library size and gene length, making expression levels comparable across samples. Now, run the script “**htseq_to_TPM_conversion.py**”, which first divides the read counts for each gene by its length in kilobases to produce Reads Per Kilobase (RPK) values. These values are then scaled so that the sum of all RPKs in a sample equals one million. After that, run the R script “**PCA_using_TPM_data.R**” to create a PCA, PCC, and scree plot.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_htseq/tpm$ python3 htseq_to_TPM_conversion.py
Data loaded successfully. Here are the first few rows:
Gene_ID Gene_length SRR15276765 SRR15276766 SRR15276767 SRR15276773
0 PF3D7_0100100 7.617 7 91 106 77
1 PF3D7_0100200 1.226 9 25 10 15
2 PF3D7_0100300 4.141 0 0 4 0
3 PF3D7_0100400 1.274 1 58 95 114
4 PF3D7_0100500 0.112 3 7 9 11
Calculated TPM values:
Gene_ID Gene_length TPM_SRR15276765 TPM_SRR15276766 TPM_SRR15276767 TPM_SRR15276773
0 PF3D7_0100100 7.617 6.972948 13.690640 13.699433 12.579384
1 PF3D7_0100200 1.226 55.699892 23.367694 8.029532 15.224863
2 PF3D7_0100300 4.141 0.000000 0.000000 0.950901 0.000000
3 PF3D7_0100400 1.274 5.955701 52.170486 73.406559 111.349438
4 PF3D7_0100500 0.112 203.238296 71.621982 79.105226 122.215775
```

```
cut -d',' -f3-6 tpm_results.txt > tpm_for_pca.txt
```

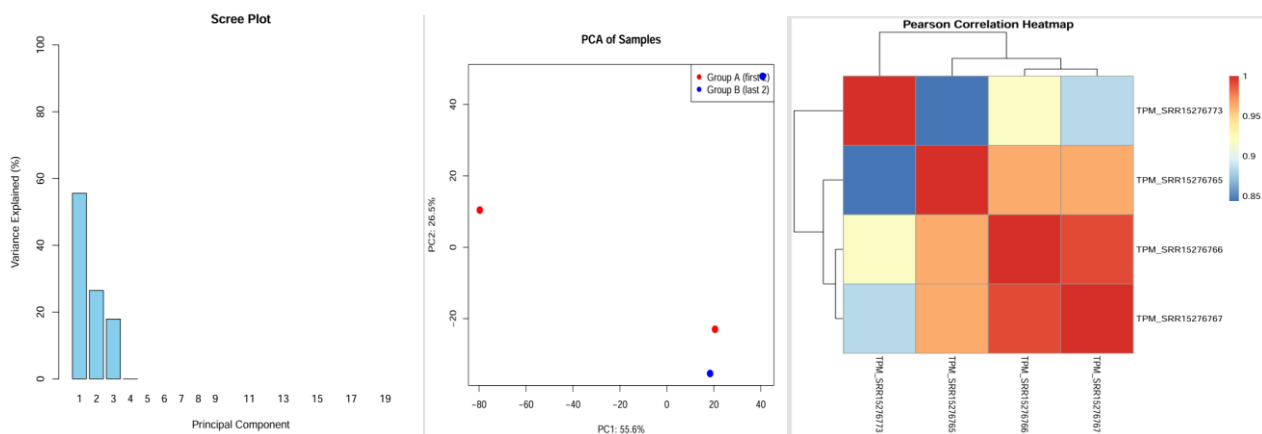
```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_htseq/tpm$ head -5 tpm_for_pca.txt
TPM_SRR15276765,TPM_SRR15276766,TPM_SRR15276767,TPM_SRR15276773
6.972947548100187,13.690640137534045,13.699433220127144,12.579383974172323
55.69989201408904,23.367693994495813,8.029531752109056,15.224862989863128
0.0,0.0,0.9509013212350353,0.0
5.955700995052604,52.17048617144768,73.4065591183785,111.34943751518733
```

nano PCA_using_TPM_data.R

```
# 1. Load data
res <- read.table("tpm_for_pca.txt", header = TRUE, sep = ",", check.names = FALSE)

# ---- PCA Scatter ----
# Define groups: first sample = Group A, others = Group B
# Define groups: first 2 samples = Group A, last 2 samples = Group B
groups <- c(rep("Group A", 2), rep("Group B", 2))
```

Rscript PCA_using_TPM_data.R

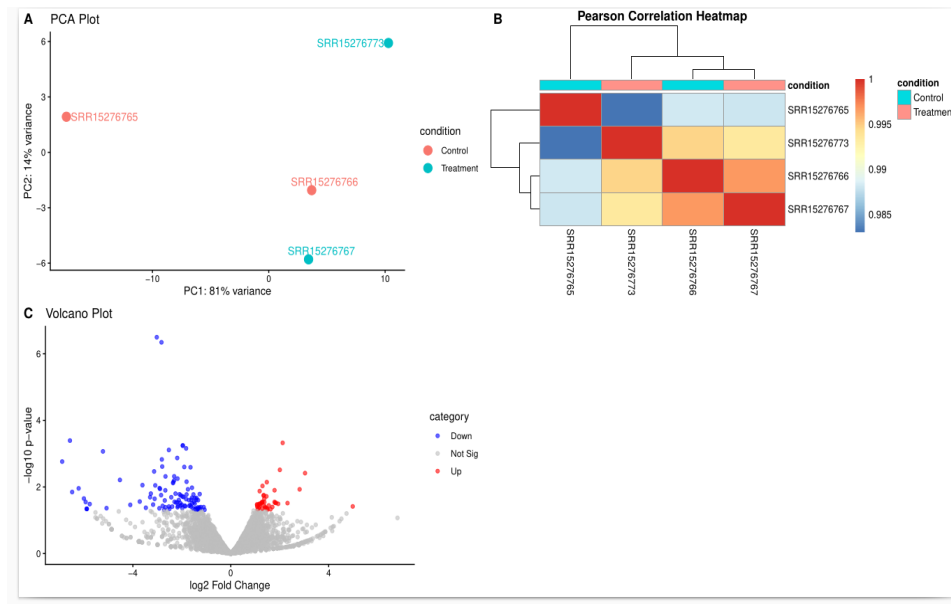


- Run the differential expression analysis using all the three methods: Deseq2, limma and edgeR and check the cutoff values for fold change and generate images.

DESeq2 takes raw RNA-seq read count data as input and applies a negative binomial distribution model with shrinkage estimators for dispersion and fold change. The default normalization method is the rlog transformation, and the output includes differentially expressed genes with log2 fold changes, p-values, and FDR-adjusted significance values.

Rscript Deseq2_analysis.R

	A	B	C	D	E	F	G
		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
1							
2	PF3D7_0100100	49.6244600566783	0.458178036980425	0.74838391691695	0.612223254166043	0.54039005815153	0.997200049557361
3	PF3D7_0100200	16.5266122364639	-1.48031413701265	1.11069524051492	-1.3327815615077	0.182603498313156	0.997200049557361
4	PF3D7_0100300	0.545078838799255	1.98475760064084	4.84911657503365	0.409302925580226	0.682317368811034	0.997200049557361
5	PF3D7_0100400	44.582351081628	1.66209400045848	0.898328356459314	1.85020765347927	0.0642836262956832	0.997200049557361
6	PF3D7_0100500	7.23301326977768	-0.105435601448177	1.53758633722211	-0.0685721503214336	0.945330187494533	0.997200049557361
7	PF3D7_0100600	2.76210596299397	-0.885400211416191	2.65501981364729	-0.333481583401025	0.738770788950509	0.997200049557361



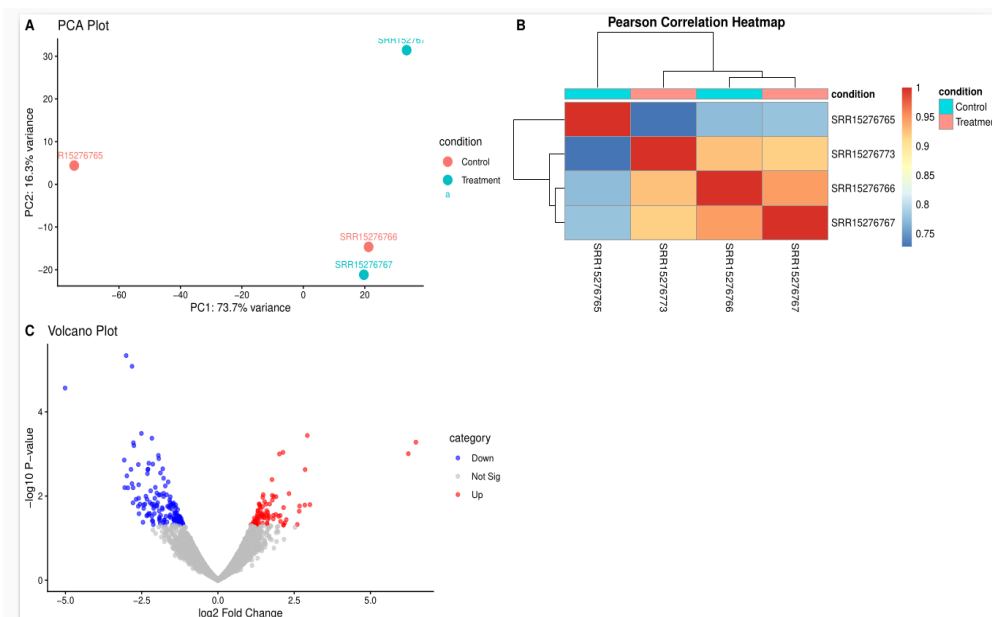
```
awk -F ',' ' $3 >= 0.5 && $6 < 0.05 ' DESeq2_results.csv > deseq2_upregulated.csv
```

```
awk -F ',' ' $3 <= -0.5 && $6 < 0.05 ' DESeq2_results.csv > deseq2_downregulated.csv
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_htseq/DeSeq2$ wc -l deseq2_upregulated.csv
41 deseq2_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_htseq/DeSeq2$ wc -l deseq2_downregulated.csv
106 deseq2_downregulated.csv
```

EdgeR also uses raw count data and models gene expression with the negative binomial distribution, but it emphasizes empirical Bayes methods for estimating dispersion parameters. The results provide ranked lists of differentially expressed genes with $\log_2$ fold changes and statistical significance measures such as p-values and FDR.

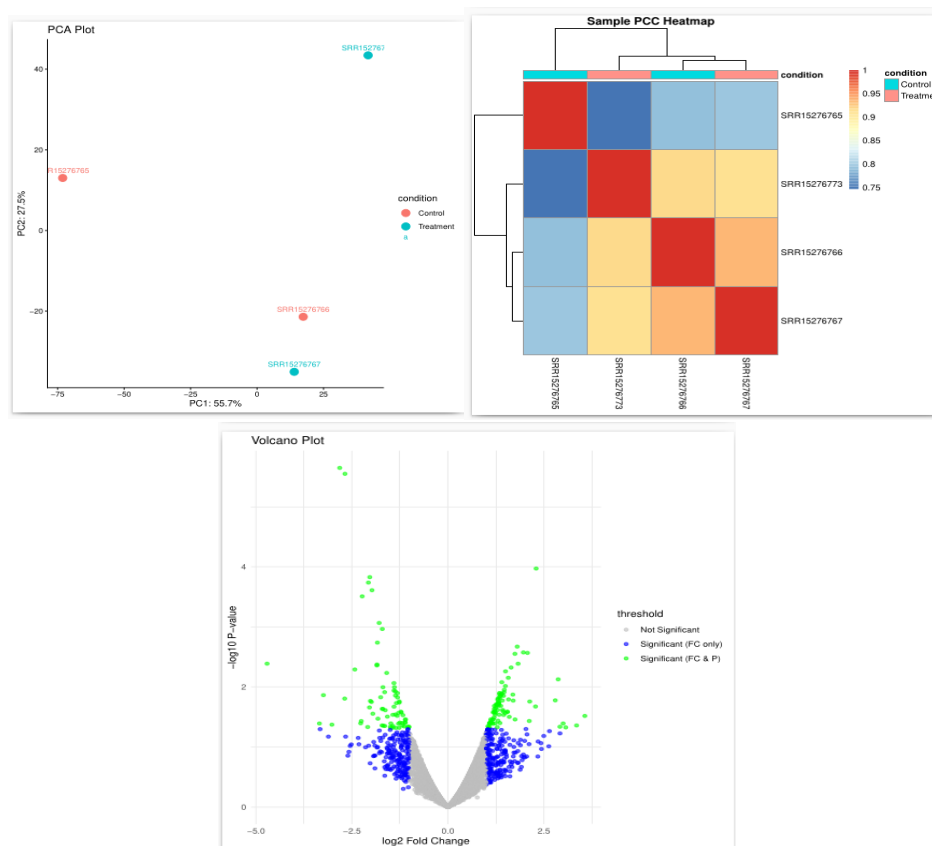
Rscript edgeR.R



```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_htseq/edgeR$ awk -F ',' ' '$2>=0.5 && $5<0.05' edgeR_results.csv > edgeR_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_htseq/edgeR$ awk -F ',' ' '$2<=-0.5 && $5<0.05' edgeR_results.csv > edgeR_downregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_htseq/edgeR$ wc -l edgeR_upregulated.csv
85 edgeR_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_htseq/edgeR$ wc -l edgeR_downregulated.csv
139 edgeR_downregulated.csv
```

Another method in differential expression analysis using the **limma method**, paired with the voom transformation for RNA-seq data, works on normalized expression values derived from counts or microarray intensities. The output consists of differentially expressed genes reported with log fold changes, moderated t-statistics, and adjusted p-values.

Rscript limma.R



For FeatureCount data:

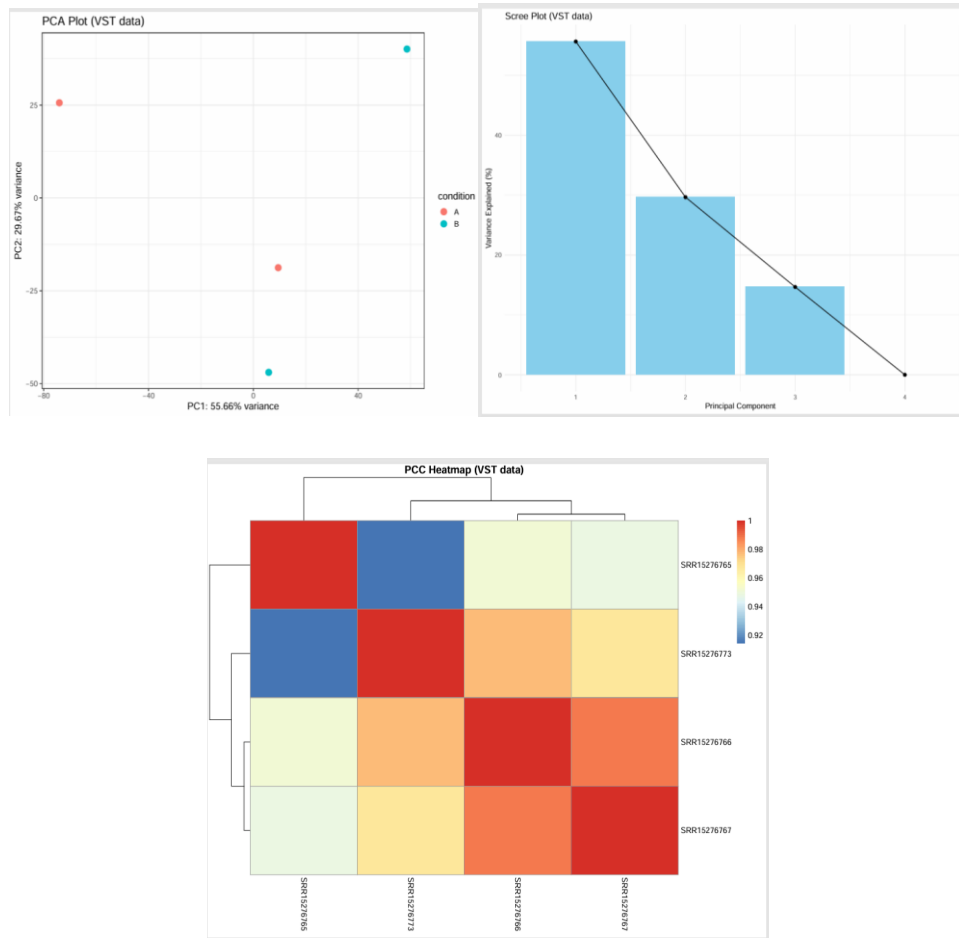
- Generate PCA, and PCC plots using vst normalisation.

nano PCA_PCC_using_vst_log_data.R

```
# -----
# 1. Import read count matrix
# -----
counts <- read.table("~/home/ibab/NGS/assignment_11/for_HISAT2/for_feature_count/raw_reads_hisat_feature_count.txt", header = TRUE, row.names = 1, sep = "\t")

# Sample info
samples <- data.frame(
  row.names = colnames(counts),
  condition = c("A", "A", "B", "B") # adjust groups
)
```

Rscript PCA_PCC_using_vst_log_data.R



- **Generate PCA and PCC plots using TPM normalisation.**

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_feature_count/tpm$ ls
SRR15276765.txt SRR15276766.txt SRR15276767.txt SRR15276773.txt raw_reads_hisat_feature_count.txt
```

Command	Purpose
<code>grep -P "^PF3D7" SRR15276765.txt awk 'BEGIN{OFS="\t"} {print \$1, \$6/1000}' > pf3d7_length_kb.txt</code>	Select only PF3D7 rows from the featureCounts file and convert the Length column from base pairs to kilobases.
<code>sort -k1,1 pf3d7_length_kb.txt > pf3d7_length_kb.sorted</code> <code>sort -k1,1 raw_reads.txt > raw_reads.sorted</code>	join requires files to be sorted on the key column (Geneid).
<code>tail -n +2 raw_reads.sorted > raw_reads.noheader</code>	Remove the header so it doesn't interfere with join.
<code>awk '{ \$1=\$1 } 1' OFS="\t" raw_reads.noheader > raw_reads.tab</code>	Ensure the raw reads file is tab-delimited (join requires consistent delimiters).

<pre>join -t \$'\t' -1 1 -2 1 pf3d7_length_kb.sorted raw_reads.tab awk 'BEGIN{OFS="\t"} {print \$1,\$2,\$3,\$4,\$5,\$6}' > merged_raw_reads_with_length.txt</pre>	Join the gene length file and raw reads file on Geneid, keeping Geneid, Length(kb), and sample counts.
<pre>(echo -e "Geneid\tLength\tSRR15276765\tSRR15276766\tSRR152767 67\tSRR15276773"; cat merged_raw_reads_with_length.txt) > tmp && mv tmp merged_raw_reads_with_length.txt</pre>	Prepend a proper header line to the merged file.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_feature_count/tpm$ head -5 merged_raw_reads_with_length.txt
Geneid Length SRR15276765 SRR15276766 SRR15276767 SRR15276773
PF3D7_0100100 6.492 138 957 952 791
PF3D7_0100200 0.996 55 222 75 122
PF3D7_0100300 3.984 4 6 6 2
PF3D7_0100400 1.119 12 65 107 122
```

```
GNU nano 6.2 htseq_to_TPM_conversion.py *
import pandas as pd

# Load the data
file_path = 'merged_raw_reads_with_length.txt' # Update the file path as needed
data = pd.read_csv(file_path, sep='\t')
```

python3 htseq_to_TPM_conversion.py

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_feature_count/tpm$ python3 htseq_to_TPM_conversion.py
Data loaded successfully. Here are the first few rows:
  Geneid Length SRR15276765 SRR15276766 SRR15276767 SRR15276773
0 PF3D7_0100100 6.492      138      957      952      791
1 PF3D7_0100200 0.996       55      222       75      122
2 PF3D7_0100300 3.984        4        6        6        2
3 PF3D7_0100400 1.119       12       65      107      122
4 PF3D7_0100500 0.112        4        6       12       14
Calculated TPM values:
  Geneid Length TPM_SRR15276765 TPM_SRR15276766 TPM_SRR15276767 TPM_SRR15276773
0 PF3D7_0100100 6.492    7.993798    11.852839    10.270511     8.449422
1 PF3D7_0100200 0.996   20.766149    17.921841     5.273945     8.494337
2 PF3D7_0100300 3.984     0.377566     0.121094     0.105479     0.034813
3 PF3D7_0100400 1.119     4.032773     4.670595     6.697108     7.560643
4 PF3D7_0100500 0.112   13.430574     4.307469     7.504070     8.668401
```

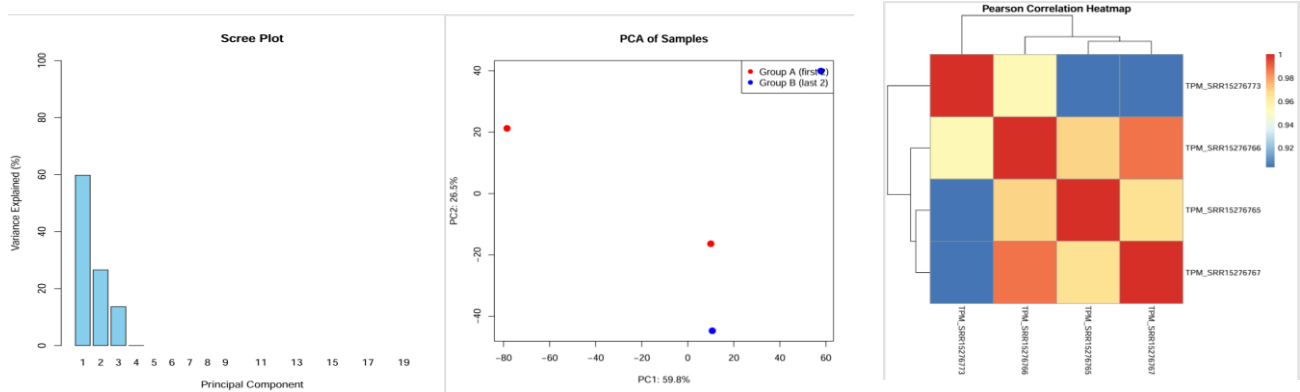
cut -d',' -f3-6 tpm_results.txt > tpm_for_pca.txt

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_HISAT2/for_feature_count/tpm$ head -5 tpm_for_pca.txt
TPM_SRR15276765,TPM_SRR15276766,TPM_SRR15276767,TPM_SRR15276773
7.993798342282865,11.852839464481393,10.270511472843063,8.449421521711471
20.766148915193533,17.92184065872448,5.273944555667299,8.494337186053958
0.3775663439126096,0.12109351796435461,0.10547889111334599,0.034812857319893274
4.032772960181868,4.6705953845768935,6.6971083608766815,7.560643286246419
```

```
GNU nano 6.2 PCA_using_TPM_data.R *
# =====
# PCA + Scree Plot + PCC (Base R, no tidyverse)
# =====

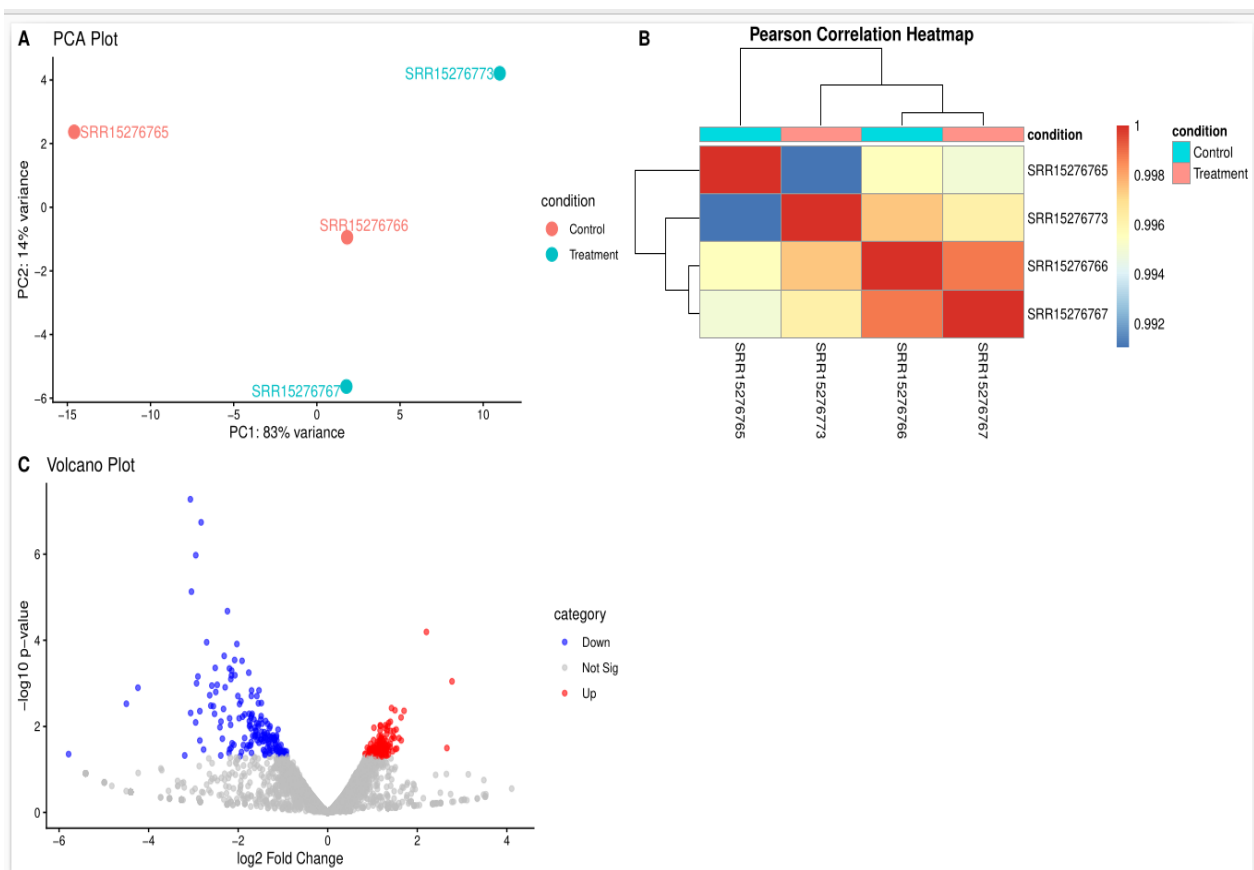
# 1. Load data
res <- read.table("tpm_for_pca.txt", header = TRUE, sep = ",", check.names = FALSE)
```


Rscript PCA_using_TPM_data.R



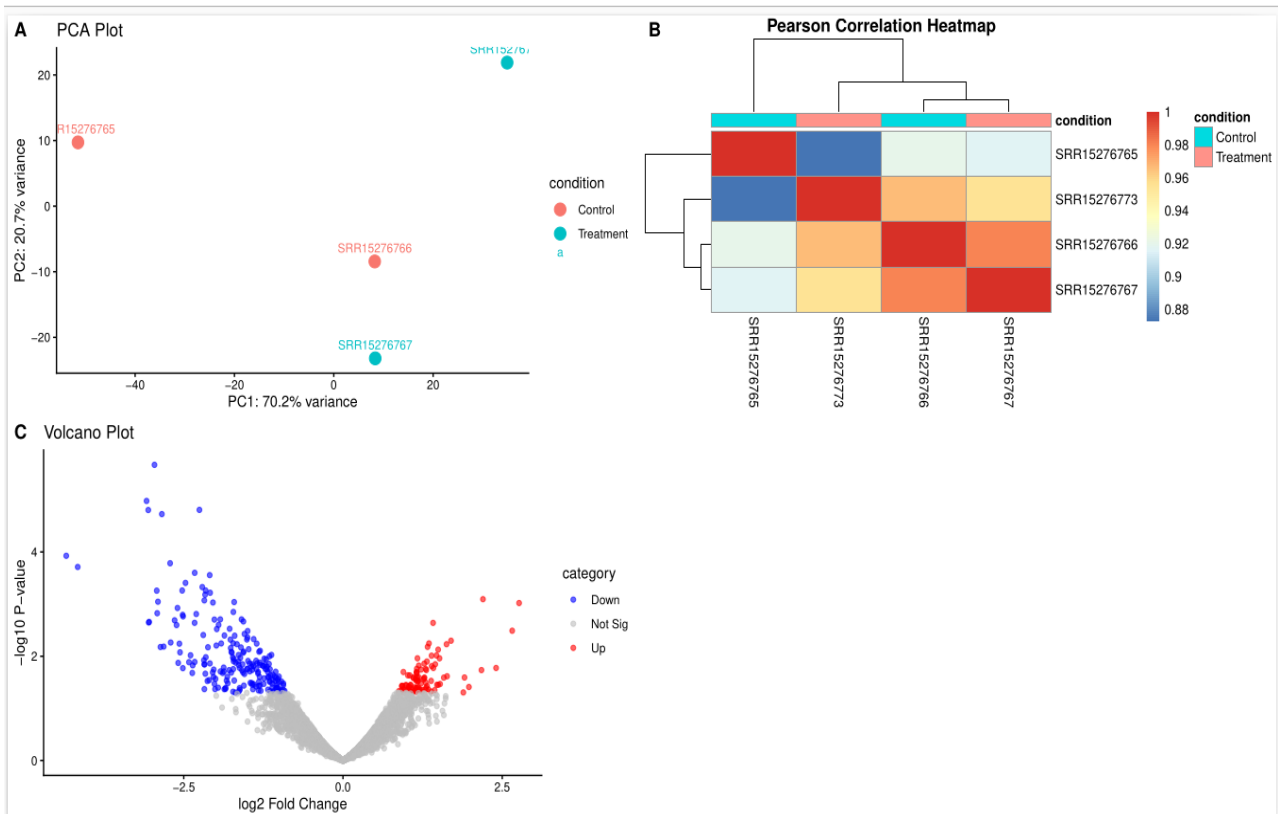
- Run the differential expression analysis using all the three methods: Deseq2, limma and edgeR and check the cutoff values for fold change and generate images.

Rscript Deseq2_analysis.R



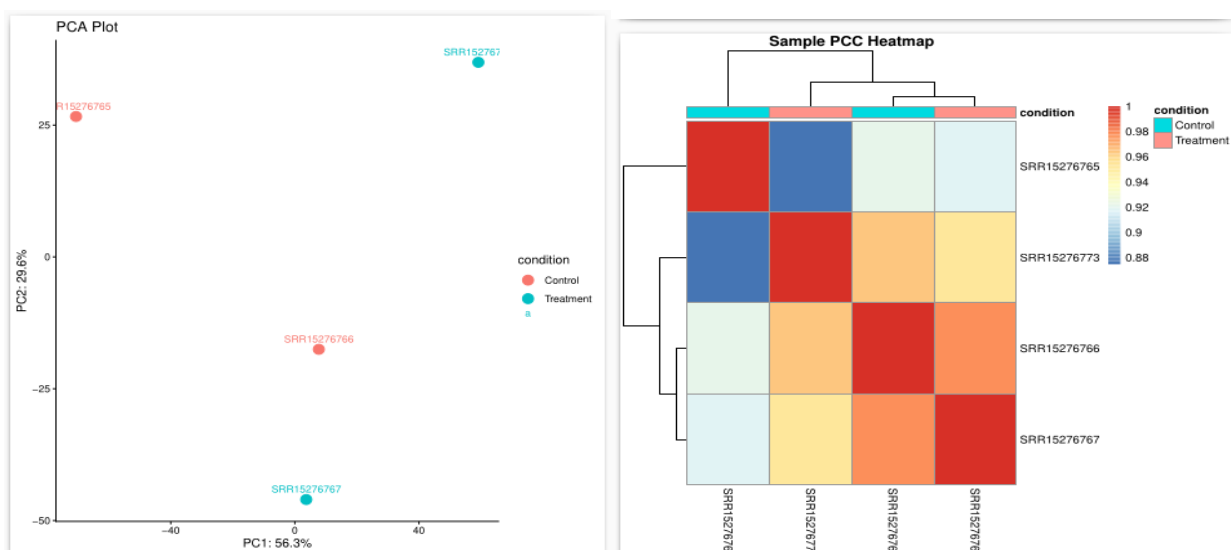
```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/Deseq2$ awk -F ',' ' $3>=0.5 && $6<0.05' DESeq2_results.csv > deseq2_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/Deseq2$ awk -F ',' ' $3<=-0.5 && $6<0.05' DESeq2_results.csv > deseq2_downregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/Deseq2$ wc -l deseq2_upregulated.csv
126 deseq2_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/Deseq2$ wc -l deseq2_downregulated.csv
165 deseq2_downregulated.csv
```

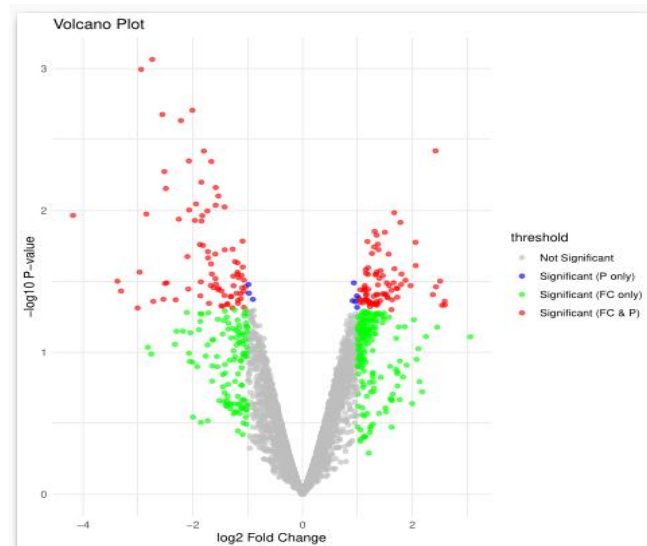
Rscript edgeR.R



```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/edgeR$ awk -F ',' ' $2>=0.5 && $5<0.05' edgeR_results.csv > edgeR_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/edgeR$ awk -F ',' ' $2<=-0.5 && $5<0.05' edgeR_results.csv > edgeR_downregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/edgeR$ wc -l edgeR_upregulated.csv
96 edgeR_upregulated.csv
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/assignment_11/for_HISAT2/for_feature_count/edgeR$ wc -l edgeR_downregulated.csv
203 edgeR_downregulated.csv
```

Rscript limma.R



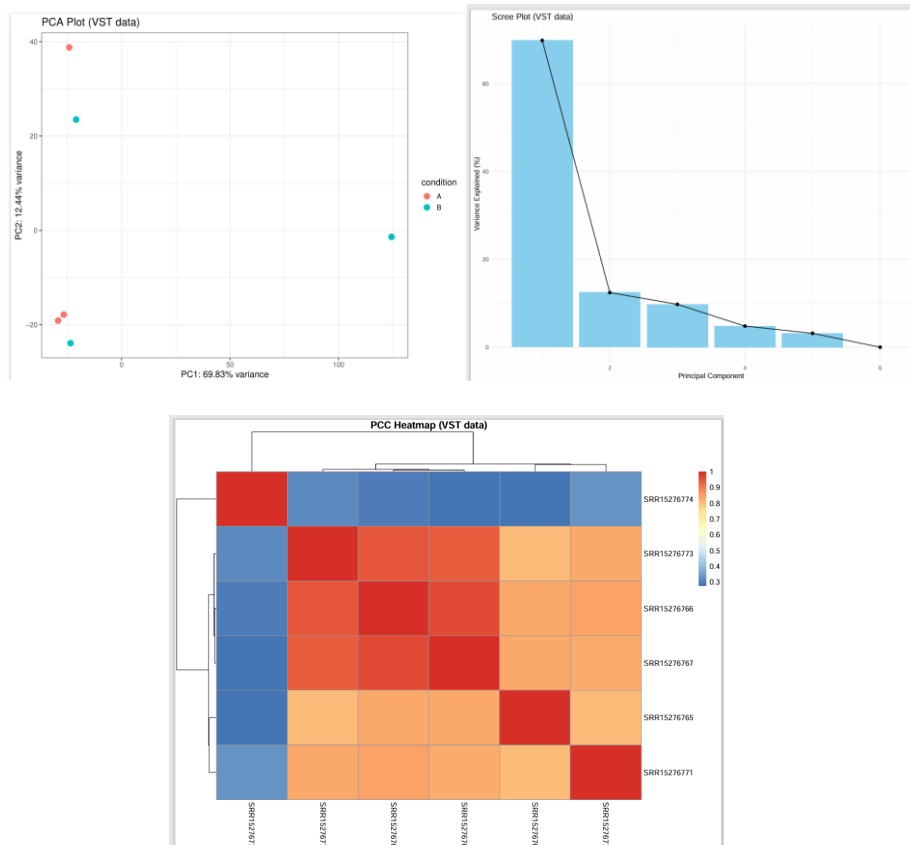


Exercise 2: Read Count with STAR Alignments

For htseq data:

- Generate PCA, and PCC plots using vst normalisation.

Rscript PCA_PCC_using_vst_log_data.R



- **Generate PCA and PCC plots using TPM normalisation.**

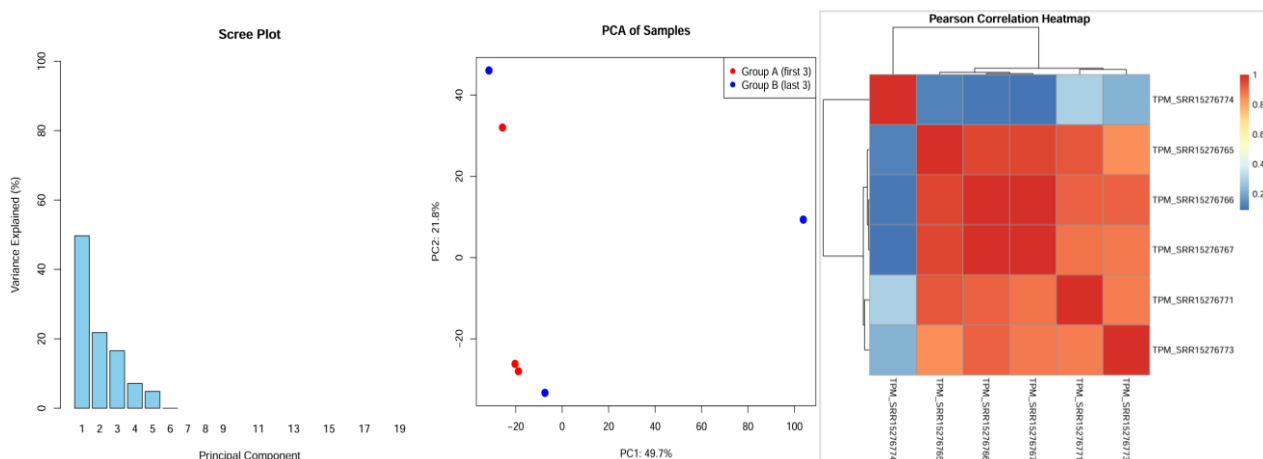
Same commands were followed here also as done in HISAT2 – for extracting the length column from .gtf file, converting it into kb, and then merging into raw reads .txt file.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/tpm$ head -5 raw_reads_with_length_reordered.txt
Gene_ID Gene_length SRR15276765 SRR15276766 SRR15276767 SRR15276771 SRR15276773 SRR15276774
PF3D7_0100100 7.617 8 95 110 8 76 12
PF3D7_0100200 1.226 9 24 10 15 0
PF3D7_0100300 4.141 0 0 4 0 0
PF3D7_0100400 1.274 1 59 95 24 116 74
```

python3 htseq_to_TPM_conversion.py

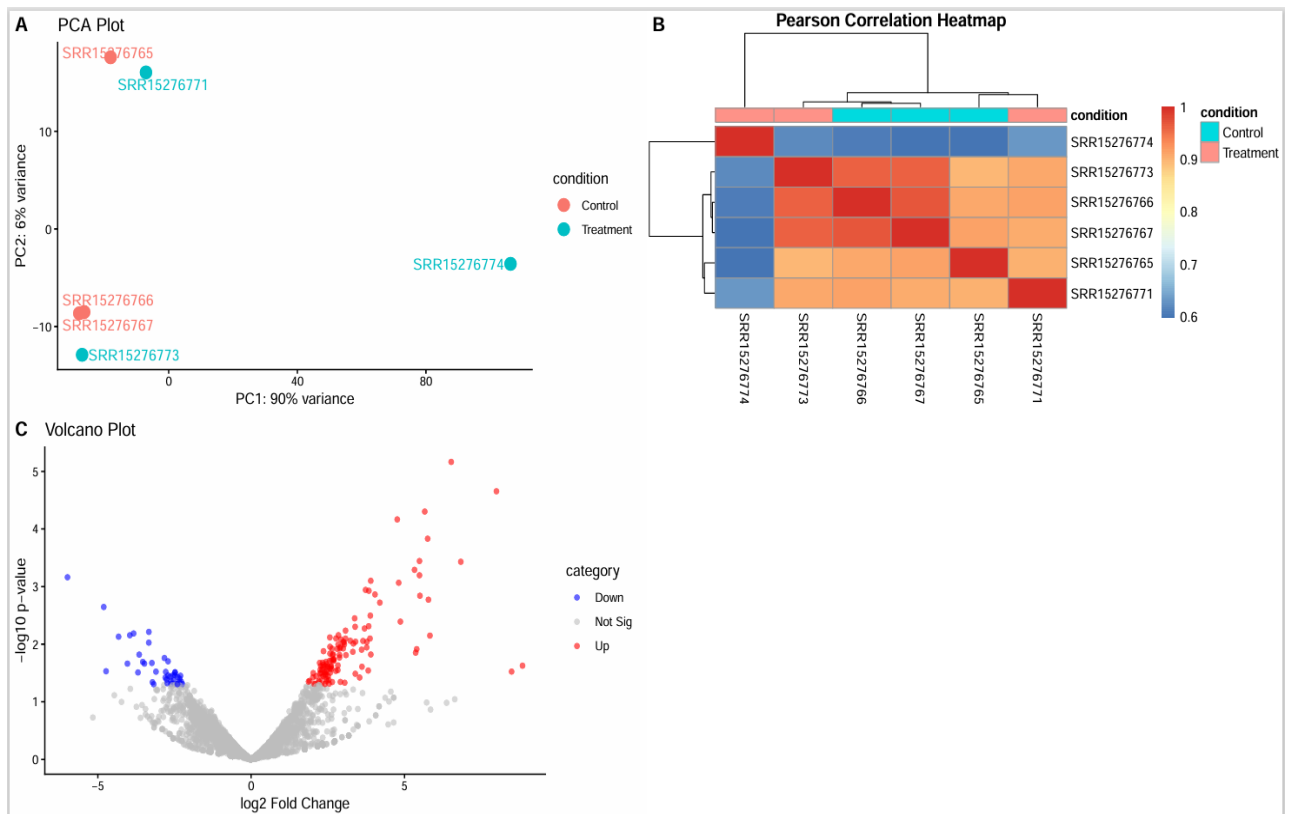
```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/tpm$ python3 htseq_to_TPM_conversion.py
Data loaded successfully. Here are the first few rows:
Gene_ID Gene_length SRR15276765 SRR15276766 SRR15276767 SRR15276771 SRR15276773 SRR15276774
0 PF3D7_0100100 7.617 8 95 110 8 76 12
1 PF3D7_0100200 1.226 9 24 10 15 0
2 PF3D7_0100300 4.141 0 0 4 0 0
3 PF3D7_0100400 1.274 1 59 95 24 116 74
4 PF3D7_0100500 0.112 3 6 12 0 12 0
Calculated TPM values:
Gene_ID Gene_length TPM_SRR15276765 ... TPM_SRR15276771 TPM_SRR15276773 TPM_SRR15276774
0 PF3D7_0100100 7.617 7.934920 ... 6.089342 12.066407 32.029214
1 PF3D7_0100200 1.226 55.461111 ... 47.290497 14.796163 0.000000
2 PF3D7_0100300 4.141 0.000000 ... 0.000000 0.000000 0.000000
3 PF3D7_0100400 1.274 5.930169 ... 109.221004 110.112566 1180.895009
4 PF3D7_0100500 0.112 202.367031 ... 0.000000 129.572114 0.000000
[5 rows x 8 columns]
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/tpm$ cut -d',' -f3-8 tpm_results.txt > tpm_for_pca.txt
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/tpm$ head -5 tpm_for_pca.txt
TPM_SRR15276765,TPM_SRR15276766,TPM_SRR15276767,TPM_SRR15276771,TPM_SRR15276773,TPM_SRR15276774
7.934920144286127,13.953746878704552,13.897798314580482,6.089342215117618,12.066406694894065,32.029214349129404
55.46111140408307,21.901404304990315,7.8495869614533245,47.29049719876723,14.796163142065781,0.0
0.0,0.0,0.9295912692336898,0.0,0.0,0.0
5.930169421019174,51.81240773800266,71.76149084776048,109.22100389140712,110.112566060284,1180.89500926227
```

Rscript PCA_using_TPM_data.R



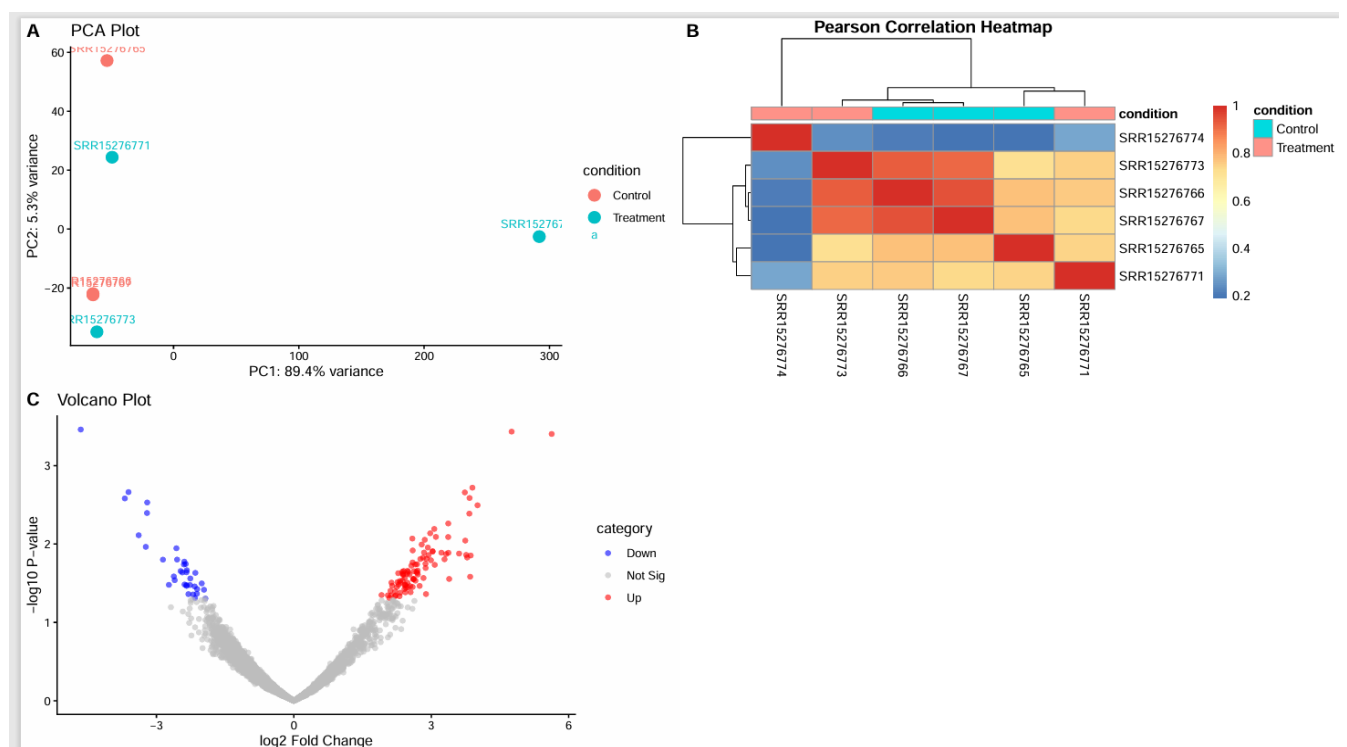
- **Run the differential expression analysis using all the three methods: Deseq2, limma and edgeR and check the cutoff values for fold change and generate images.**

Rscript Deseq2_analysis.R



```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/Deseq2$ awk -F ' ' '$3>=0.5 && $6<0.05' DESeq2_results.csv > deseq2_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/Deseq2$ awk -F ' ' '$3<=-0.5 && $6<0.05' DESeq2_results.csv
> deseq2_downregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/Deseq2$ wc -l deseq2_upregulated.csv
118 deseq2_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/Deseq2$ wc -l deseq2_downregulated.csv
42 deseq2_downregulated.csv
```

Rscript edgeR.R

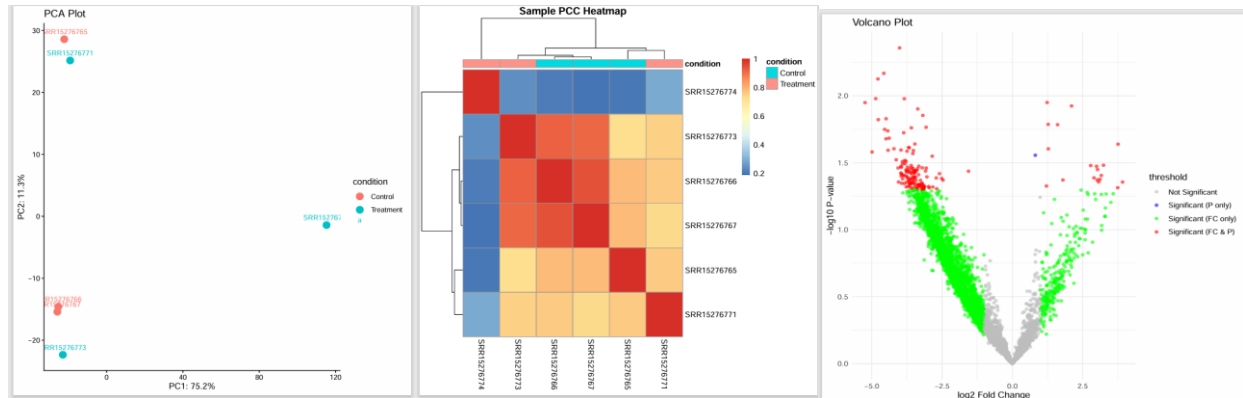


```

ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/edgeR$ awk -F ' ' '$2>=0.5 && $5<0.05' edgeR_results.csv > edgeR_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/edgeR$ awk -F ' ' '$2<=-0.5 && $5<0.05' edgeR_results.csv > edgeR_downregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/edgeR$ wc -l edgeR_upregulated.csv
95 edgeR_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_htseq/edgeR$ wc -l edgeR_downregulated.csv
36 edgeR_downregulated.csv

```

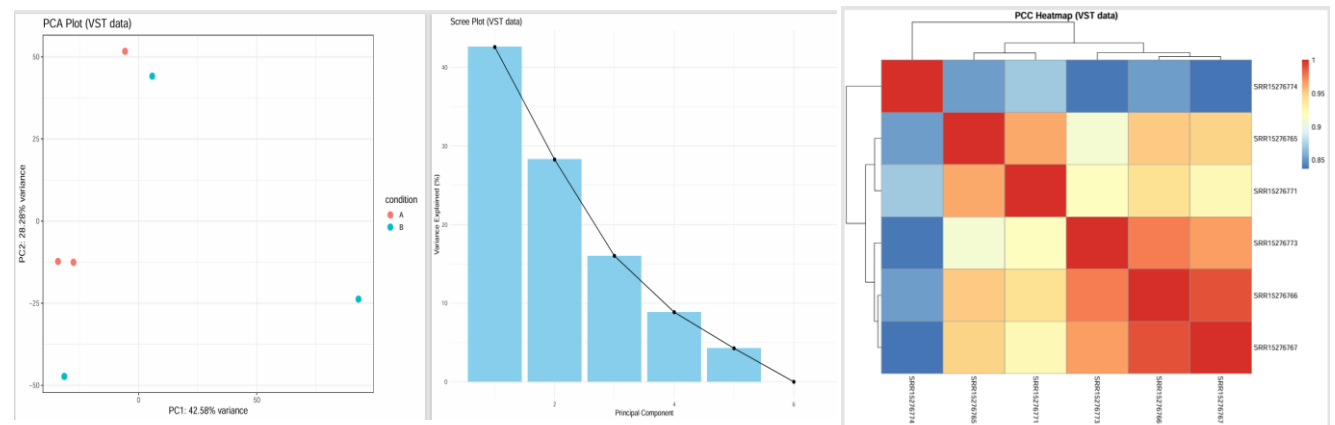
Rscript limma.R



For FeatureCount data:

- **Generate PCA, and PCC plots using vst normalisation.**

Rscript PCA_PCC_using_vst_log_data.R



- **Generate PCA and PCC plots using TPM normalisation.**

Same commands were followed here also as done in HISAT2 – for extracting the length column from the FeatureCount output file (.txt), converting it into kb, and then merging into raw reads .txt file.

```

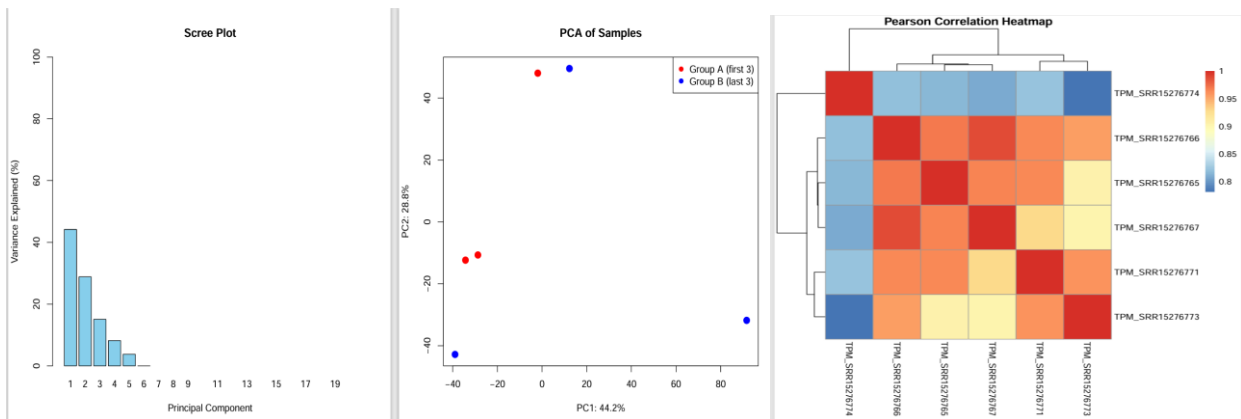
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/tpm$ head -5 merged_raw_reads_with_length.txt
Geneid Length SRR15276765 SRR15276766 SRR15276767 SRR15276771 SRR15276773 SRR15276774
PF3D7_0100100 6.492 145 956 945 207 783 97
PF3D7_0100200 0.996 56 218 75 59 122 44
PF3D7_0100300 3.984 4 6 6 4 2 0
PF3D7_0100400 1.119 12 65 107 30 124 74

```

python3 featurecount_to_TPM_conversion.py

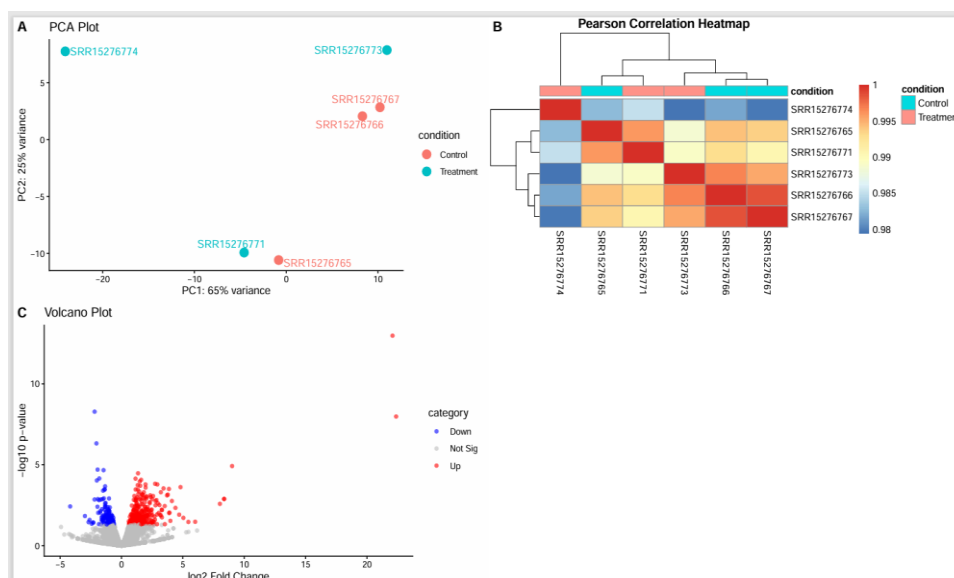
```
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/tpm$ python3 featurecount_to_TPM_conversion.py
Data loaded successfully. Here are the first few rows:
  Geneid  Length  SRR15276765  SRR15276766  SRR15276767  SRR15276771  SRR15276773  SRR15276774
0 PF3D7_0100100  6.492      145         956         945         207         783         97
1 PF3D7_0100200  0.996       56         218         75         59         122        44
2 PF3D7_0100300  3.984       4           6           6           4           2           0
3 PF3D7_0100400  1.119      12          65         107         30         124        74
4 PF3D7_0100500  0.112       4           7          12           0           12           0
Calculated TPM values:
  Geneid  Length  TPM_SRR15276765  TPM_SRR15276766  TPM_SRR15276767  TPM_SRR15276771  TPM_SRR15276773  TPM_SRR15276774
0 PF3D7_0100100  6.492      8.370671       11.711834       10.080865       7.902412       8.248406       8.422241
1 PF3D7_0100200  0.996     21.071695     17.407752      5.214905     14.681165     8.376976     24.901631
2 PF3D7_0100300  3.984      0.376280      0.119778      0.104298      0.248833      0.034332      0.000000
3 PF3D7_0100400  1.119      4.019036      4.619860      6.622138      6.644450      7.578415     37.276582
4 PF3D7_0100500  0.112     13.384827      4.970791      7.420066      0.000000      7.327402      0.000000
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/tpm$ cut -d',' -f3-8 tpm_results.txt > tpm_for_pca.txt
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/tpm$ head -3 tpm_for_pca.txt
TPM_SRR15276765,TPM_SRR15276766,TPM_SRR15276767,TPM_SRR15276771,TPM_SRR15276773,TPM_SRR15276774
8.370670975680914,11.711834021093493,10.080865371687453,7.902412393422321,8.248406413718143,8.422241157983873
21.071694879869465,17.407751685987346,5.214905494437668,14.681165309682036,8.376976413211851,24.901630670059387
```

Rscript PCA_using_TPM_data.R



- Run the differential expression analysis using all the three methods: Deseq2, limma and edgeR and check the cutoff values for fold change and generate images.

Rscript Deseq2_analysis.R



```

ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/Deseq2$ wc -l deseq2_upregulated.csv
321 deseq2_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/Deseq2$ wc -l deseq2_downregulated.csv
139 deseq2_downregulated.csv

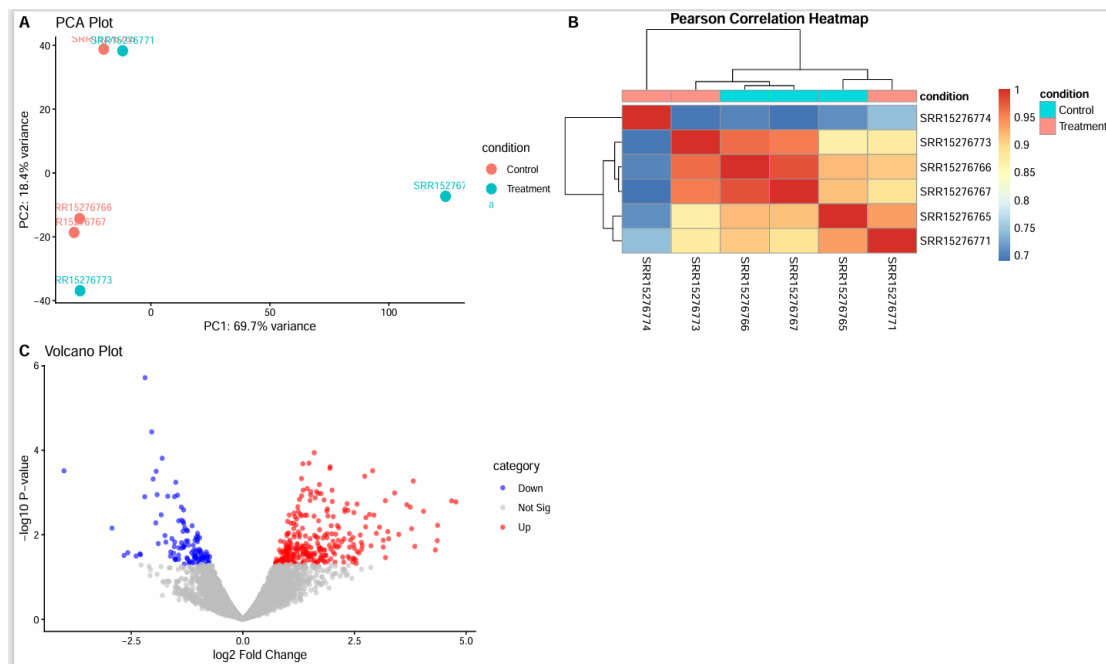
```

Rscript edgeR.R

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/edgeR$ awk -F ',' '$2>=0.5 && $5<0.05' edgeR_results.csv > edgeR_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/edgeR$ awk -F ',' '$2<=-0.5 && $5<0.05' edgeR_results.csv > edgeR_downregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/edgeR$ wc -l edgeR_upregulated.csv
274 edgeR_upregulated.csv
ibab@LAPTOP-BVSTVK8Q:~/NGS/assignment_11/for_STAR/for_feature_count/edgeR$ wc -l edgeR_downregulated.csv
111 edgeR_downregulated.csv

```



Rscript limma.R

